

IBT COLLEGE OF CANADA

LABORATORY REPORT

Configuring Router-on-a-Stick Inter-VLAN Routing

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1.0 Objective

The objective of this laboratory exercise was to configure inter-VLAN routing using a "Router-on-a-Stick" approach, enabling communication between devices situated in different VLANs (VLAN 10 and VLAN 30) through a single physical router interface (G0/0) connected to a trunk port (G0/1) on switch S1.

2.0 Summary & Understanding (100–150 Words)

In this lab, I learned how to bridge communication between isolated network segments using a single router interface configured with 802.1Q subinterfaces. Initially, devices in different VLANs (such as PC1 in VLAN 10 and PC3 in VLAN 30) could not communicate because traffic cannot cross VLAN boundaries without a Layer 3 device. By configuring subinterfaces (G0/0.10 and G0/0.30) on router R1 with appropriate IP addresses and encapsulation types, and enabling trunking on the switch port connecting to the router (G0/1), I successfully established inter-VLAN routing. Verifying the configuration with ping tests confirmed 100% successful packet delivery, demonstrating how network segmentation and efficient routing are implemented in enterprise environments.

3.0 Screenshots & Verification Section

The screenshot displays a network simulation environment. The top section shows a topology diagram with a router (R1) connected to a switch (S1). PC1 is connected to S1 via F0/11 (VLAN 10, 172.17.10.0/24). PC3 is connected to S1 via F0/6 (VLAN 30, 172.17.30.0/24). The router R1 has a G0/0 interface connected to S1's G1/1 interface. The bottom section shows a packet capture table with the following data:

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
●	Successful	PC1	PC3	ICMP	■	0.000	N	0	(edit)	(delete)
●	Successful	PC1	PC3	ICMP	■	0.000	N	1	(edit)	(delete)
●	Successful	PC1	PC1	ICMP	■	0.000	N	2	(edit)	(delete)